

Java Arithmetic Operators

The Foundation of Mathematical Logic in Code

Arithmetic operators are used to perform common mathematical operations. In Java, these range from basic addition to specialized operations like finding remainders.

1. Quick Reference Table

Op	Name	Description	Example
+	Addition	Adds two values together	$10 + 5 = 15$
-	Subtraction	Subtracts one value from another	$10 - 5 = 5$
*	Multiplication	Multiplies two values	$10 * 5 = 50$
/	Division	Divides one value by another	$10 / 5 = 2$
%	Modulus	Returns the division remainder	$10 \% 3 = 1$
++	Increment	Increases value by 1	<code>x++</code>
--	Decrement	Decreases value by 1	<code>x--</code>



Pro Tip: In integer division (e.g., $10 / 3$), Java will discard the decimal and return 3. Use `double` if you need the exact decimal value!

2. Practical Implementation

```
int x = 10;
int y = 3;

System.out.println(x + y); // Output: 13
System.out.println(x * y); // Output: 30
System.out.println(x % y); // Output: 1 (Remainder of 10/3)

int z = 5;
++z; // Increment z to 6
System.out.println(z); // Output: 6
```

Summary Recap

- **Standard Math:** Use `+`, `-`, `*`, `/` for general calculations.
- **The Modulus (%):** Extremely useful for checking if a number is even or odd (e.g., `num % 2 == 0`).
- **Shorthand:** `++` and `--` are vital for loops and simple counters.
- **Integer Bound:** Be careful with division results when using `int` variables.